NEWTON SCHOOL

PRE-CALCULUS | ANSWER KEY v2

Polar Coordinates: Concepts, Conversion & Graph Sketching

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Part A ??Why & Formulas

A1. Why polar is bad for squares, ideal for circles

> Why are polar coords inefficient for a square's corners but ideal for a circle's points?

Conceptual | Coordinate System Choice

A polar point is described by *one* angle + *one* distance from the pole ??exactly the two quantities that stay constant or change predictably on a circle (distance fixed, angle sweeps).

On a square, both the distance to the center and the direction vary *non-smoothly* as you walk along the edges. Each corner has the same distance to the center as some other corner but a totally different angle, and the side lengths require ugly mixed expressions in r and θ (a square has no clean polar equation). Rectangular (x, y) is the natural choice for grid-aligned shapes.

Polar = good for radial / circular motion; rect = good for grid shapes.

A2. Derive r^2 and $\tan\theta$

From $x = r\cos\theta$ and $y = r\sin\theta$, derive both r^2 and $\tan\theta$ in terms of x, y.

Pythagorean Identity | Derivation

For r^2: Square and add:

$$x^2 + y^2 = (r\cos\theta)^2 + (r\sin\theta)^2 = r^2(\cos^2\theta + \sin^2\theta) = r^2 \cdot 1 = r^2.$$

So $\boxed{r^2 = x^2 + y^2.}$

For $\tan\theta$: Divide y by x (assuming $r\cos\theta \neq 0$):

$$\frac{y}{x} = \frac{r\sin\theta}{r\cos\theta} = \tan\theta.$$

So $\boxed{\tan\theta = \frac{y}{x}.}$

$r^2 = x^2 + y^2, \quad \tan\theta = y/x$ (with quadrant check).

A3. TRUE / FALSE with justification

(a) $(r,\theta) \equiv (r, \theta + 2\pi)$ (b) $(-r,\theta) \equiv (r, \theta - \pi)$ (c) $(0,\theta)$ is the pole for any θ (d) $(r,\theta) \equiv (-r, -\theta)$

Polar Equivalence

(a) TRUE. Adding 2π is a full rotation ??same ray, same distance.

(b) TRUE. Reflecting through the pole flips the radius sign and shifts the angle by π (either $+\pi$ or $-\pi$ works).

(c) TRUE. If $r = 0$ the point is at distance 0 from the pole = the pole itself, regardless of θ.

(d) FALSE in general. Counter-example: $(r,\theta) = (1, \pi/4)$ sits at $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$. But $(-r,-\theta) = (-1, -\pi/4)$: angle $-\pi/4$ points to $(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$; since $r = -1$, flip to opposite side $\Rightarrow (-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$. Different from the original.

a-T, b-T, c-T, d-F

Part B ??Plotting & Alternative Reps

B-Worked Example (plot $(-2, 5\pi/6)$ + two alt reps) ??solution shown in worksheet.

B1. Plot $(3,\ -7\pi/4)$, identify quadrant

Plot the polar point and state the quadrant.

Negative Angle | Plotting

Reduce angle. $-\dfrac{7\pi}{4}+2\pi=\dfrac{\pi}{4}$. So $(3,\ -7\pi/4)\equiv(3,\ \pi/4)$.

Plot. $r=3$ along the ray at $\theta=\pi/4$ (45째 above positive x-axis).

Quadrant. $\pi/4$ ray sits in **Q-I**.

Quadrant I, at $(3\cos\pi/4,\ 3\sin\pi/4)=\left(\frac{3\sqrt{2}}{2},\ \frac{3\sqrt{2}}{2}\right)$.

B2. Plot $(-4,\ 13\pi/6)$ + equivalent with $r>0,\ 0\le\theta<2\pi$

Plot, then find equivalent representation.

Negative r | Angle Reduction

Reduce angle. $\dfrac{13\pi}{6}-2\pi=\dfrac{\pi}{6}$. So $(-4,\ 13\pi/6)\equiv(-4,\ \pi/6)$.

Flip for $r>0$. Apply $(-r,\theta)\equiv(r,\theta+\pi)$: $(-4,\ \pi/6)\equiv(4,\ \pi/6+\pi)=(4,\ 7\pi/6)$.

Plot. Ray at $\theta=7\pi/6$ (Q-III, 210째), distance 4 from pole.

$(4,\ 7\pi/6)$ **??Q-III**

B3. $(5,\ 7\pi/6)$ with $r<0$ AND $-2\pi<\theta<0$

Find an equivalent rep with both $r<0$ and θ negative.

Alternative Rep

Flip sign of r. $(5,\ 7\pi/6)\equiv(-5,\ 7\pi/6-\pi)=(-5,\ \pi/6)$. But $\pi/6>0$??doesn't satisfy $\theta<0$.

Subtract 2π. $\pi/6-2\pi=-\dfrac{11\pi}{6}$. Check: $-2\pi<-\dfrac{11\pi}{6}<0$ ✓.

$$\left(-5,\ -\frac{11\pi}{6}\right)$$

B4. Plot four points; which two coincide?

> $(2, \pi/3),\ (-2, \pi/3),\ (2, -\pi/3),\ (-2, -\pi/3)$. Which two coincide?

Polar Equivalence

Convert each to rectangular $(r\cos\theta,\ r\sin\theta)$:

- $(2, \pi/3) \to (1,\ \sqrt{3})$??Q-I.
- $(-2, \pi/3) \to (-1,\ -\sqrt{3})$??Q-III.
- $(2, -\pi/3) \to (1,\ -\sqrt{3})$??Q-IV.
- $(-2, -\pi/3) \to (-1,\ \sqrt{3})$??Q-II.

All four are **distinct** ??none coincide. They form a rectangle centered at the origin (the four corners $(\pm 1,\ \pm\sqrt{3})$).

None coincide ??4 distinct points forming a rectangle.

Part C ??Point Conversion

C-Worked Example (convert $(-3, 3\sqrt{3})$ rect $\to$ polar) ??solution shown in worksheet.

C1. Polar $\to$ Rect: $(-4,\ 23\pi/4)$

> Reduce angle, then convert.

Polar?㈱ect | Angle Reduction

Reduce θ: $\dfrac{23\pi}{4} - 2\cdot 2\pi = \dfrac{23\pi}{4} - \dfrac{16\pi}{4} = \dfrac{7\pi}{4}$.

So we convert $(-4,\ 7\pi/4)$.

$x = -4\cos(7\pi/4) = -4\cdot\dfrac{\sqrt{2}}{2} = -2\sqrt{2}$.

$y = -4\sin(7\pi/4) = -4\cdot(-\dfrac{\sqrt{2}}{2}) = 2\sqrt{2}$.

$\boxed{(-2\sqrt{2},\ 2\sqrt{2})}$ **??Q-II**

C2. Polar $\to$ Rect: $(6,\ 11\pi/6)$

> Convert to rectangular.

Polar?㈱ect

$\cos(11\pi/6) = \cos(-\pi/6) = \dfrac{\sqrt{3}}{2}$. $\sin(11\pi/6) = -\dfrac{1}{2}$.

$x = 6\cdot\dfrac{\sqrt{3}}{2} = 3\sqrt{3}$.

$y = 6\cdot(-\dfrac{1}{2}) = -3$.

$\boxed{(3\sqrt{3},\ -3)}$ **??Q-IV**

C3. Rect $\to$ Polar: $(-\sqrt{6},\ -\sqrt{2})$

> Tricky Q-III. Exact form, $r > 0,\ 0 \le \theta < 2\pi$.

Rect?垦olar | Quadrant III

Quadrant. $x < 0, y < 0 \Rightarrow$ Q-III.

Radius. $r = \sqrt{(-\sqrt{6})^2 + (-\sqrt{2})^2} = \sqrt{6+2} = \sqrt{8} = 2\sqrt{2}$.

Reference angle. $\tan\theta_{ref} = \left|\frac{-\sqrt{2}}{-\sqrt{6}}\right| = \frac{\sqrt{2}}{\sqrt{6}} = \frac{1}{\sqrt{3}} \Rightarrow \theta_{ref} = \frac{\pi}{6}$.

Q-III adjust. $\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$.

$$\boxed{\left(2\sqrt{2},\ \frac{7\pi}{6}\right)}$$

C4. Rect $\to$ Polar: $(0,\ -5)$

> Edge case on the negative y-axis.

Edge Case | Axis Point

Don't divide by zero. $x = 0$ means $\tan\theta = y/x$ is undefined ??read the angle directly from the geometry.

The point $(0, -5)$ lies on the negative y-axis: distance 5 from origin, direction $\theta = 3\pi/2$.

$r = 5,\quad \theta = \frac{3\pi}{2}$.

$$\boxed{\left(5,\ \frac{3\pi}{2}\right)}$$

Part D ??Sketching $r = a\cos\theta$ / $r = a\sin\theta$

D-Worked Example (sketch $r = -2\cos\theta$) ??solution shown in worksheet.

D1. Sketch $r = 3\sin\theta$

> Use the two-panel Cartesian-trick. Identify the circle's location and diameter.

Circle (sin) | Cartesian-trick

Step 1 ??Cartesian $y = 3\sin x$: standard sine wave, amplitude 3. Zero at $x = 0$, max +3 at $x = \pi/2$, zero at $x = \pi$, min −3 at $x = 3\pi/2$, zero at $x = 2\pi$.

Step 2 ??Read the values into polar:

- $\theta = 0$: $r = 0$ (pole).
- $\theta = \pi/2$: $r = 3$??point $(0, 3)$.
- $\theta = \pi$: $r = 0$ (pole again).
- $\theta = 3\pi/2$: $r = -3$??opposite ray of $\theta = 3\pi/2$ gives $\theta = \pi/2$, distance 3 $\Rightarrow$ point $(0, 3)$ again.

Step 3 ??Connect: a circle of diameter 3 sitting on the **positive y-axis**, center $(0, 1.5)$, radius 1.5. Traced once over $\theta \in [0, \pi]$; the second half $[\pi, 2\pi]$ retraces it.

Circle, diameter 3, on positive y-axis, center $(0, 1.5)$.

D2. Sketch $r = -4\cos\theta$

> Two-panel format. Which side of x-axis is the circle on? What is the diameter?

Circle (negative cos) | Cartesian-trick

Step 1 ??Cartesian $y = -4\cos x$: flipped cosine. Min −4 at $x = 0$, zero at $\pi/2$, max +4 at π, zero at $3\pi/2$, min −4 at 2π.

Step 2 ??Read into polar:

- $\theta = 0$: $r = -4$??opposite ray of $\theta = 0$ is $\theta = \pi$, distance 4 $\Rightarrow$ point $(-4, 0)$.
- $\theta = \pi/2$: $r = 0$ (pole).
- $\theta = \pi$: $r = 4$??angle 180쨘, distance 4 $\Rightarrow$ point $(-4, 0)$ again.
- $\theta = 3\pi/2$: $r = 0$ (pole).

Step 3 ??Connect: a circle of diameter 4 sitting on the **negative x-axis**, center $(-2, 0)$, radius 2. Compare with $r = 4\cos\theta$ (positive x-axis) ??the sign of a flips the side.

Diameter = 4. Negative x-axis (left side). Center $(-2, 0)$.

Part E ??Equation Conversion

E-Worked Example (convert $y = x^2$ rect $\to$ polar) ??solution shown in worksheet.

E1. Rect $\to$ Polar: $x^2 + y^2 = 9$

Convert with $r > 0$.

Circle | Rect?뫅olar

Substitute $x^2 + y^2 = r^2$:

$r^2 = 9 \;\Rightarrow\; r = 3$ (since $r > 0$).

$\boxed{r = 3 \text{ (circle of radius 3 centered at the pole)}}$

E2. Rect $\to$ Polar: $x^2 - y^2 = 1$

Use a double-angle identity.

Hyperbola | Double-Angle

Substitute $x = r\cos\theta,\ y = r\sin\theta$:

$$(r\cos\theta)^2 - (r\sin\theta)^2 = 1 \;\Rightarrow\; r^2(\cos^2\theta - \sin^2\theta) = 1.$$

Use $\cos^2\theta - \sin^2\theta = \cos 2\theta$:

$$r^2\cos 2\theta = 1.$$

$\boxed{r^2 = \frac{1}{\cos 2\theta} = \sec 2\theta \text{ (defined where } \cos 2\theta > 0\text{).}}$

E3. Polar → Rect: $r = \dfrac{1}{\sin\theta - \cos\theta}$

Convert to rectangular form.

Polar?暠ect Line

Multiply both sides by $(\sin\theta - \cos\theta)$:

$$r(\sin\theta - \cos\theta) = 1.$$

Distribute:

$$r\sin\theta - r\cos\theta = 1.$$

Substitute $r\sin\theta = y$ and $r\cos\theta = x$:

$$y - x = 1.$$

$\boxed{y = x + 1 \text{ (a line)}}$

E4. Polar → Rect: $r^2 = \tan\theta$

Convert to rectangular form. Verify the pole.

Polar?暠ect Manipulation

Write $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$:

$$r^2 = \frac{\sin\theta}{\cos\theta}.$$

Multiply both sides by $\cos\theta$:

$$r^2\cos\theta = \sin\theta.$$

Multiply both sides by r (introduces x and y):

$$r^3\cos\theta = r\sin\theta \;\Rightarrow\; r^2\cdot(r\cos\theta) = (r\sin\theta) \;\Rightarrow\; (x^2+y^2)\,x = y.$$

Final form: $(x^2+y^2)\,x = y,$ equivalently $x^3 + xy^2 = y$.

Verify pole: $(0,0)$ gives $0 = 0\,\checkmark$. The pole is on the curve, consistent with $\theta = 0,\ r = 0$ in the original equation.

$\boxed{x^3 + xy^2 = y \text{ (equivalently } (x^2+y^2)x = y)}$